TEXORA · LECTURE SHEET

Lubrication

Two metal surfaces in contact without a lubricant film wear, heat and seize. Lubrication is the cheapest maintenance a mill does — and the most neglected. This sheet covers lubricant types, the properties that decide selection, the functions a lubricant performs, and the systems that deliver it, with textile-machine selection guidance.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 7 · LEVEL 4 TERM 1

1. What lubrication is

Lubrication is the introduction of a substance — the **lubricant** — between surfaces in relative motion to separate them wholly or partly, so that friction, wear, heat and corrosion are reduced. Depending on film thickness the regime is **hydrodynamic** (surfaces fully separated by a pressurised film, as in a well-fed journal bearing), **boundary** (film only a few molecules thick on asperities, as at slow, heavily loaded contacts) or **mixed**. Lubricant choice and delivery must match the regime: a thin oil cannot protect a slow, loaded gear; a heavy grease cannot feed a high-speed spindle.

2. Lubricant types

- **Liquid oils:** mineral oils refined from petroleum (the mill standard), synthetic oils (polyalphaolefins, esters — wider temperature range, longer life, higher cost), and vegetable oils (biodegradable niches). Additives give anti-wear, anti-oxidant, rust-inhibitor, detergent and extreme-pressure (EP) performance.
- **Greases:** oil held in a soap or synthetic thickener — a semi-solid that stays where it is put. Used in sealed or slow bearings, open gears' enclosed cases and where relubrication is infrequent. Consistency graded by NLGI number.
- **Solid lubricants:** graphite and molybdenum disulfide (MoS_2) — for very high temperature, vacuum or one-time assembly lubrication where liquids fail.
- **Gases:** air bearings in special high-speed equipment — rare in mills, listed for completeness.

Textile-specific warning: lubricants near fibre and fabric must be chosen and applied to avoid **oil staining** — a major cloth defect. Low-staining, washable (scourable) oils are specified for knitting and for loom parts over the warp, and quantity control matters as much as grade.

3. General properties

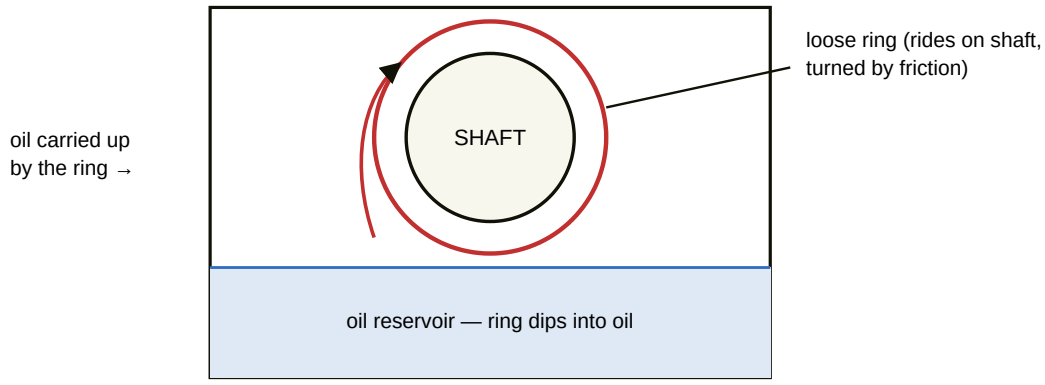
PROPERTY	MEANING	WHY IT MATTERS
Viscosity	Resistance to flow — the oil's body; graded ISO VG (mm²/s at 40 °C)	Decides film thickness: too thin → wear; too thick → drag and heat
Viscosity index (VI)	How little viscosity changes with temperature (high VI = stable)	Machines that start cold and run hot need high VI
Flash point	Temperature at which vapour first ignites momentarily	Safety margin near hot surfaces; low flash = fire risk
Fire point	Temperature at which vapour burns continuously	Used with flash point for hot-duty selection
Pour point	Lowest temperature at which the oil still flows	Cold stores, winter start-up, outdoor plant
Oiliness / film strength	Ability to cling and protect under boundary conditions	Slow, loaded, start-stop contacts
Oxidation stability	Resistance to thickening and sludge in hot, aerated service	Long drain intervals; compressor and turbine oils
Demulsibility	Ability to separate from water	Wet mill environments; steam/water contamination
Acidity / TAN	Total acid number — rises as oil oxidises in service	Condition monitoring: rising TAN signals oil change

4. Functions of a lubricant

1. **Reduce friction** — lower power loss and heat generation.
2. **Reduce wear** — protect surfaces, extend machine life (Topic 9).
3. **Carry away heat** — circulating oil cools bearings and compressors.
4. **Prevent corrosion** — film excludes moisture and air from metal.
5. **Seal** — oil film seals piston rings and screw-compressor rotors.
6. **Flush contaminants** — circulating systems carry wear particles to filters.
7. **Transmit power** — in hydraulic systems the same oil is the working fluid.

5. Lubrication systems

- **Manual / hand oiling:** oil can to cups and nipples on a schedule — simple, operator-dependent; the classic source of both missed points and over-oiling stains.
- **Splash lubrication:** rotating parts dip into an oil bath and splash oil to bearings and gears (gearboxes, engine crankcases). Needs correct oil level — too low starves, too high churns and overheats.
- **Ring oiling:** a loose ring rides on the shaft, dips into the reservoir and carries oil up to the journal continuously (Figure 1) — self-acting for horizontal shaft bearings, e.g. older line-shaft and motor bearings.
- **Wick / pad feed:** capillary feed from a reservoir — steady micro-feed for spindles and slow bearings.
- **Pressure-feed (forced) lubrication:** a pump delivers filtered oil under pressure to many points (large engines, turbines, compressors), with coolers, filters and pressure alarms.
- **Centralised systems:** one pump and metering units feed dozens of points on a machine or line, on a timer — single-line, dual-line and progressive types for grease or oil. Guarantees every point receives its measured dose; standard on modern spinning and weaving installations.
- **Oil-mist lubrication:** atomised oil in air piped to bearings — continuous micro-feed, used on high-speed spindle groups in some plants.



Ring oiling: shaft rotation drags the ring, the ring lifts oil to the journal, surplus drains back.

Figure 1 — Ring-oiling system for a horizontal shaft bearing. Flow direction is shown by the red arrow: reservoir → ring → journal → back to reservoir.

6. Lubricant selection for textile machines (typical practice — always follow the machine maker's chart first)

APPLICATION	TYPICAL LUBRICANT	KEY REQUIREMENT

Ring-frame spindles	Very light spindle oil (low ISO VG)	Low drag at very high speed; metered micro-dose
Knitting machine elements	Washable (scourable) needle oil	Must scour out of fabric — non-staining
Loom gears and cams	Medium gear oil / EP oil or grease	Film strength; containment away from warp
Compressor (air)	Compressor oil, high oxidation stability	Low carbon deposit at discharge temperature
Boiler-house pumps, fan bearings	General machine oil / grease per bearing type	Demulsibility in wet, hot surroundings
Dyeing machine drives	Gear oil, sealed systems	Water resistance; no leakage into process

Management rules: store oils clean, sealed and labelled; never mix grades; follow the maker's lubrication chart (point, lubricant, quantity, interval) as a maintenance document (Topic 9); sample critical oils for condition; train oilers — most lubrication failures are wrong oil, wrong quantity or contamination, not oil quality.

Glossary

Viscosity — a fluid's resistance to flow. **Viscosity index** — measure of how little viscosity changes with temperature. **Flash point** — temperature at which vapour flashes on ignition. **Pour point** — lowest temperature at which oil flows. **Grease** — oil thickened into a semi-solid. **Hydrodynamic lubrication** — full separation of surfaces by a pressurised film. **Centralised lubrication** — one pump feeding many points through metering units. **TAN** — total acid number, an oil-ageing indicator.

Key points

• Match viscosity to speed and load: thin for fast/light, heavy or grease for slow/loaded.
• VI, flash/fire point, pour point and oxidation stability decide suitability beyond viscosity alone.
• Systems range from the oil can to pressure-feed and centralised metering — delivery reliability is the real difference.
• In textiles, staining control is a lubrication specification, not an afterthought.

Practice

1. Choose a lubrication system for (a) a large mill motor's horizontal bearing, (b) 200 points on a new spinning line, (c) a gearbox with dipping gears — and justify each in one line.
2. An oil must serve a machine that starts in a cold store at 5 °C and runs with bearings at 90 °C. Which two properties dominate selection, and why?
3. Why is over-oiling a quality fault in a knitting mill, not merely wasteful?
4. State the functions of a circulating pressure-feed oil system beyond friction reduction (name three).

Answer key

1. (a) Ring oiling — self-acting continuous feed for a horizontal journal. (b) Centralised system — guarantees measured doses to every point on schedule. (c) Splash — gears dip and distribute oil inside the closed case.
2. Pour point (must still flow at 5 °C) and viscosity index (must not thin excessively by 90 °C — high VI keeps the film at running temperature while remaining fluid when cold).
3. Excess needle oil transfers to yarn and fabric as oil stains, which survive as visible defects and may not scour out — a direct cloth-quality and claims issue.
4. Any three of: cooling the bearings, flushing wear particles to the filter, corrosion protection, sealing, and (in engines) carrying additives to all contacts continuously.
